Measuring the Diversity a Feed Actually Exposes

A standard, its attacks, and what it takes to verify it

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Abstract

Knowing what a recommender algorithm actually exposes requires a quantity one can observe without access to the platform's code. This note proposes one — the **Exposed Diversity Index** (EDI), the entropy of the items served, projected onto a declared catalogue of viewpoints and weighted by the attention each rank commands — and then subjects it to the attacks it must withstand to serve as a standard.

Three of those attacks succeed against the naive formulations, and each forces a correction. A floor bearing on a feed's *labels* is saturated at zero cost: a platform able to decouple label from content obtains full marks for strictly zero content diversity. Rao's quadratic entropy, the first replacement considered, has a *bimodal* constrained optimum: it would prescribe the very polarisation it claims to measure. And any rank-blind measure is satisfied by *burying* the divergent items — a platform certified at 0.70 exposes only 0.36 of real diversity, and closing the loophole doubles the engagement cost.

Verifying such a standard on real data requires knowing *exposure*. We give three checks to run on a recommender log — does its order say anything, is there enough to estimate, and in what form — and apply them to three public logs. MIND does not preserve the served order, and the usual estimator returns five incompatible severities there; Baidu-ULTR does preserve it; the Open Bandit Dataset permits this work's only confrontation with a ground truth, where the value of a never-deployed policy is recovered to within 2.5% against 31.6% error for the naive estimate.

The principal conclusion of this part is however negative for the method we had just built. Baidu-ULTR publishes a display-time column allowing exposure to be **measured** rather than estimated through clicks: severity is then 0.88 ± 0.05 over 143 documents, not 1.09 ± 0.09 over 55 — the estimate overstates decay by 23%, because a click conflates examination with attractiveness. Direct measurement further refutes the cascade model on that log, by two independent routes, and shows the $R^{-\eta}$ law to be the worst of three fits on the observed curve. We had finally concluded that a rich display format placed above removes 8.4 points of examination from what follows, and that the rank weighting was therefore a property of the **page served** rather than of rank. A matched measurement withdraws that conclusion: with the *item* held fixed only 6% remains (risk ratio 0.940, 95% CI $[0.916, 0.964]$), two thirds of the published effect being a composition of queries. Its bearing on the index is 0.0025, against 0.0350 for the $1/R$ convention: the rank law is therefore **portable**, and what threatens the measurement is not the page but not having measured.

A fourth log, EB-NeRD, finally carries both expected columns — the served list and a declared section. The first check says its **recorded order does not contain the rank** ($z = +1.05$, where the log would have detected a severity one hundred and thirty-four times smaller). The rank-blind index is therefore measured for the first time on real feeds — 0.50 per user-day, 5.1 sections equally served out of 26 — and the exposed index only *bounded*, to within 0.104: at a floor of 0.40, a third of feeds breach it whatever their ordering, and six in ten remain undecidable. What must be required of a log is therefore not the rank but the **verifiable** rank — and the exchangeability test is what verifies it. The catalogue, finally, decides the level far more than the feed does: the same corpus scores 0.50 over twenty-six sections and 0.92 over three, while the ordering between readers holds down to six modalities and then collapses.

The index came from a thermodynamic model whose every own claim the audit refuted, and which no longer appears here: the quantum decoherence analogy that started this work survives none of its transfers. All code, tests and figures are reproducible in containers.

Part I
The measurement

1 The metrological problem

Regulating a recommender algorithm presupposes an observable quantity. Counting problematic items will not do: the indicator lags, it is gameable, and it requires ruling on the substance of every item. The quantity adopted here is of a different nature — it bears on the *shape* of exposure, not on the content of messages:

> What share of the attention served to a reader goes to each of the viewpoints the platform could have served?

The question has three virtues: it can be measured from outside, it requires no judgement on the truth of any item, and it admits a threshold expressed as a percentage, hence transferable across platforms.

It also has three traps, and this note is organised around them.

1. **A label is not an item.** Measuring the diversity of *announced categories* leaves the platform to choose what is announced.
2. **A composition is not an exposure.** A feed may contain the required diversity and place it where nobody looks.
3. **A logged click is not a preference.** Observed clicks were produced by the platform's ranking, not by the one under evaluation.

Each trap is measurable, and each was measured. What survives all three is the retained definition.

2 The exposed diversity index

Given a feed of n positions, a reference catalogue of k viewpoints declared by the declarer, and an attention discount w_R attached to rank R, the *exposed* distribution is

$$q_i = \frac{\sum_{R=1}^{n} w_R \mathbb{1}[\text{the item served at rank } R \text{ falls in bin } i]}{\sum_{R=1}^{n} w_R}, \tag{1}$$

and the index is its normalised Shannon entropy:

$$\mathrm{EDI} = \frac{H(q)}{\log_2 k}, \qquad H(q) = -\sum_{i=1}^{k} q_i \log_2 q_i. \tag{2}$$

Three choices distinguish (1) from the entropy one would write spontaneously, and each follows from an attack that succeeded (§3).

Choice	Reason	Without it
q bears on the items served, projected onto the catalogue's bins	a label is chosen, an item is observed	the index reaches 1.000 for zero content diversity, at zero cost
each rank is weighted by w_R	a reader consults the first item far more often than the last	the standard is satisfied by burial: 0.70 certified, 0.36 exposed
the denominator $\log_2 k$ comes from the declared catalogue	comparing platforms requires the same unit	a closed feed shows one modality, the denominator degenerates and the index flatters

Attention discount. The results below use $w_R = 1/R$, the reciprocal-rank discount standard in ranking evaluation. The choice of w belongs to the declarer, as does the catalogue; what matters here is that it be *declared*, since it is what makes the standard non-circumventable by order.

Associated control quantity. The gap between (2) computed on the feed's composition (w_R constant) and on its exposed distribution ($w_R = 1/R$) is zero for a platform that does not relegate, and grows with burial. It is the only quantity in this work that compares a measure *with itself*, and hence the only one that thresholds without an external reference.

How to publish the figure. A normalised entropy is not a diversity: it is not linear in what we mean by "twice as diverse". Its conversion into an **effective number of viewpoints**, ${}^1D = k^{\mathrm{EDI}}$, is [Jost, 2006]: a floor of 0.70 on a catalogue of four is 2.6 effective viewpoints, and the 0.44 actually exposed by the burying feed is 1.9.

Protocol. The unit is the item *served*, not the item consulted: the platform is audited, not the reader. The index is computed per reader over a sliding window and then aggregated; the quantity of interest is the *share of the population below a threshold*, not the mean, since a satisfactory mean can hide an entirely enclosed minority.

3 The adversarial test

A standard is judged by what it forces a rational adversary to do. We therefore place the platform in the role it will occupy: maximising engagement *subject to* the constraint, rather than complying in good faith.

Method. A feed has n positions to fill from a catalogue of k viewpoints, hence k^n possible feeds. For $n = 8$ and $k = 4$, all 65,536 feeds are *enumerated*: the optimum obtained is exact, not the output of a heuristic. This is not a luxury. These are negative results — standards that fail — and an optimum missed by a solver would look exactly like a standard that holds.

3.1 A label-based standard saturates at zero cost

If the platform controls the label independently of the item, the entropy of labels stops constraining the entropy of items. Optimisation confirms it without nuance: under full decoupling the platform obtains an index of 1.000 — full marks — for strictly zero content diversity, without giving up a point of engagement. One need not go that far: at half decoupling, the constraint retains only 36% of its force.

The consequence is direct: **the measure must bear on the items served**. It calls for a complementary diagnostic, the *signature excess* — the gap between two indices computed on the same feed, relative to what an honest catalogue would show at the same index — which is zero for an honest platform and grows with decoupling.

3.2 The first replacement prescribed polarisation

Rao's quadratic entropy [Rao, 1982] — the mean distance between two items drawn from the feed, known in recommendation as *intra-list distance* — appeared to answer the objection: it sees items, not labels. Yet its engagement-constrained optimum is **bimodal**: at a given required diversity, the cheapest way to supply it is to serve two extreme blocks and empty the centre. A standard built on it would prescribe the polarisation it claims to measure. The defect is known in the diversification literature [Ohsaka and Togashi, 2023]; it is recovered here as a direct consequence of constrained optimisation.

The retained floor therefore bears on an *entropy*, computed on the items served projected onto the catalogue's bins — a measure whose constrained optimum is spread rather than bimodal.

3.3 A rank-blind standard is circumvented by burial

The measures above bear on the feed's *composition*. Retested on *ordered* feeds, all four candidate diversity measures are circumvented the same way: serve the divergent items at the last ranks, where attention no longer goes.

Measure	**Cost**	**Displayed**	**Exposed**	**Cost if rank-aware**
Rao's entropy	8.2%	0.750	**0.355**	18.9%
Position entropy	10.7%	0.774	**0.443**	20.9%
Gaussian ILD	floor unattainable			
Target proximity	5.9%	0.750	**0.628**	10.6%

The optimal feed always has the same shape: items of the preferred viewpoint on top, divergent ones relegated. **A platform certified at 0.70 exposes only 0.36.** A rank-aware floor closes the loophole and *doubles* the engagement cost: a standard that cost no more would close nothing.

The gap grows with the requirement — 0.262 at a floor of 0.40, 0.395 at 0.70 for Rao's entropy: the more diversity a rank-blind standard demands, the more profitable burial becomes.

4 Measuring exposure

Everything above presupposes knowing what attention goes to which rank. The standard model posits an examination probability decreasing with rank [Joachims et al., 2017],

$$e(R) = R^{-\eta}, \tag{3}$$

whose severity η was, in the first version of this work, *posited.* It can be estimated.

4.1 Estimation by item fixed effects

Under (3), click probability factorises as $P(\text{click} \mid i, R) = g(i)\, R^{-\eta}$, hence

$$\log \mathrm{CTR}(i, R) = \log g(i) - \eta \log R. \tag{4}$$

Relevance $g(i)$ is an item fixed effect: it is not estimated but eliminated by within-item centring. What remains is the only variation that identifies η — that of the *same item seen at different ranks.* This is the simplest form of intervention harvesting [Agarwal et al., 2019]: it requires no experiment, only that the platform did not always rank the same items in the same places.

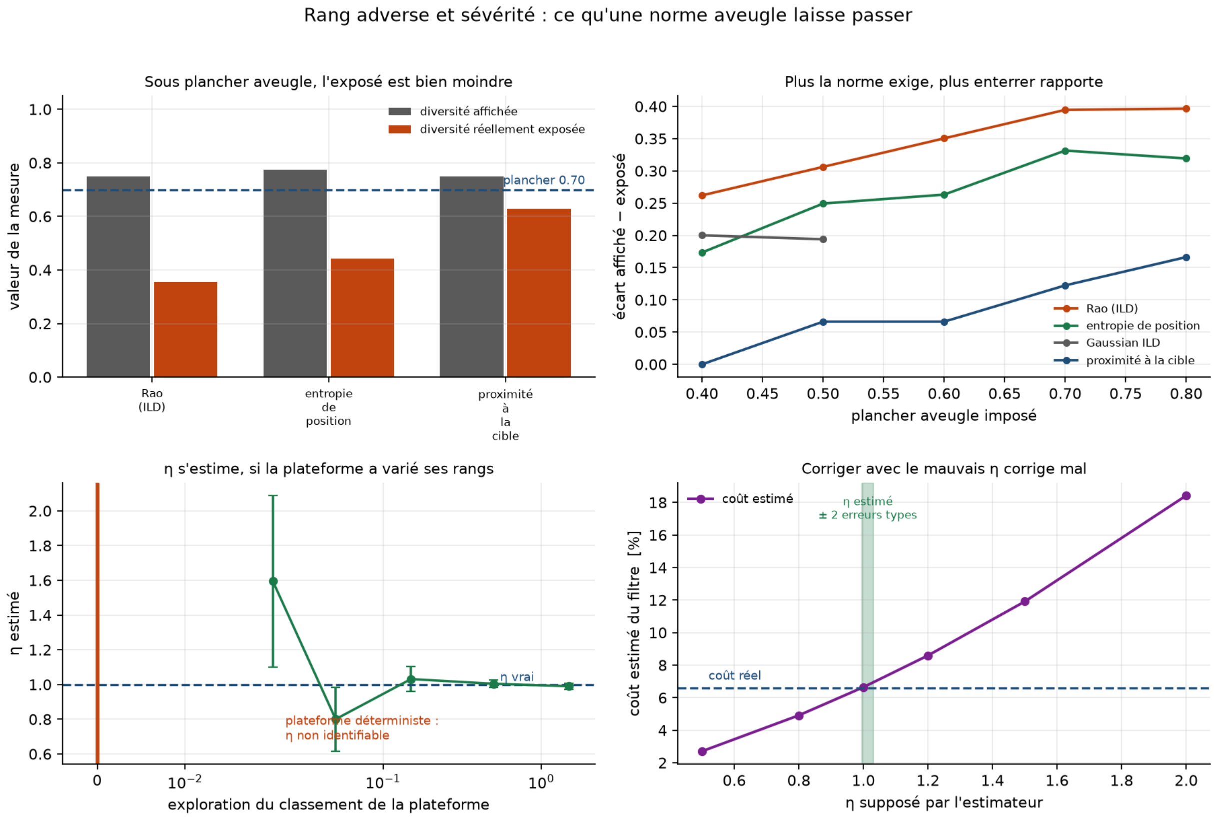

Figure 1: Displayed versus exposed diversity under a rank-blind floor; the gap by floor level; position-bias severity recovered by fixed effects, and the exploration threshold below which it ceases to be identifiable; the cost of a posited η.

On simulated data the estimator recovers η from 0.4 to 1.6. Under a deterministic policy it *refuses* to answer rather than invent a figure: no item changes rank, and the severity is not in the data.

Positing η instead of estimating it is not innocuous: where the true cost of a diversity filter is 6.6%, a posited η of 0.5 understates it by 59% and a posited η of 2.0 overstates it by 179% — the very order of magnitude of the bias one claimed to correct.

4.2 The check that must come first: exchangeability

The refusal described above catches the visible case — no rank variation — and misses the invisible one: *abundant and artificial* variation. What is needed is a check that counts not the variation but the *informativeness* of the order.

For a feed of length L carrying k clicks, let $u_R = (R - 1/2)/L$ be the normalised rank. Under the hypothesis that clicks are indifferent to position *given the feed*, the clicked positions are a draw without replacement among the L positions, with exactly known expectation and variance:

$$\mathbb{E}\Big[\sum_{\text{clicks}} u_R\Big] = k\,\bar{u}, \qquad \mathbb{V}\Big[\sum_{\text{clicks}} u_R\Big] = \frac{k(L-k)}{L-1}\,\sigma_u^2. \tag{5}$$

Conditioning on the feed immunises the test against the mixture of feed lengths, the mean quality of a feed's items, and its reader's appetite for clicking. Position bias concentrates clicks at the top: it makes the standardised deviation *negative*, so the sign of the rejection is itself a verification.

A test that rejects nothing says nothing until one knows what it would reject. Calibration is done by simulation at identical feed structure: on MIND's, the test detects $\eta = 0.02$ at twelve standard deviations, and the minimum detectable severity is $\eta \approx 0.004$.

5 Evaluating a re-ranking

Measuring a feed is one thing; estimating what a different ranking would cost is another. The available clicks were produced by the platform's policy, not by the one under evaluation.

Replay is wrong, by a considerable margin. Replaying logged clicks on a re-ranked feed is off by 201% at the median, up to 851%, and the sign of the error is not guaranteed: across sixty draws at identical configuration it overstates in fifty-six cases and understates in four. A naive figure is therefore not even an upper bound.

The correction and its price. Inverse propensity weighting is unbiased provided the logging policy is known and gives every action the target policy might choose a chance. Its price is variance, hence the usual variants — self-normalisation [Swaminathan and Joachims, 2015], weight clipping — and the diagnostic that must accompany any figure: the **effective sample size**, which states how many observations actually carry the estimate.

A confrontation with ground truth. The Open Bandit Dataset [Saito et al., 2020] contains two buckets served in parallel by two different policies, and publishes the true propensities. One can therefore estimate the value of the uniform policy *from another policy's data alone*, and compare it with its value measured where it actually served.

Estimator	**Value**	**Error**
Ground truth (measured, 452,949 impressions)	0.005124	—
Naive (observed click rate)	0.006743	+31.6%
IPS	0.005253	+**2.5**%
Self-normalised	0.004768	−7.0%
IPS clipped at 10	0.004473	−12.7%

The correction works. The diagnostic forbids celebrating it: the effective sample size is 1,513 for 4,077,727 impressions, i.e. 0.04%. The estimate is unbiased and rests on the equivalent of fifteen hundred observations; that the error lands at 2.5% owes as much to luck as to method. This is the real case that justifies the rule: publish the effective size next to the figure.

6 Three public logs

The instruments of §4 and §5 decide whether a dataset permits the measurement. Applied to three public logs, they give three different answers.

6.1 MIND: an order that is not one

The reference dataset for news recommendation [Wu et al., 2020] shows ideal rank coverage: a median item appears at sixteen distinct positions. Yet test (5) detects nothing — $z = +0.12$ ($p = 0.91$) across 156,965 feeds, replicated at $z = +0.28$ on the second split — in a dataset where it would detect $\eta = 0.02$ at twelve standard deviations. The recorded order is indistinguishable from a shuffle.

The click-rate curve by position nonetheless falls from 0.108 to 0.038 and fits $\hat{\eta} = 0.39$: a *composition artefact*, deep positions existing only in long feeds, whose click rate per item is mechanically lower. At fixed feed length the slope is zero and changes sign from one length to the next.

Finally, estimator (4) accepts the dataset, declares itself identifiable, and returns a severity that is an increasing function of a mere nuisance threshold: −0.131 at five impressions, +0.053

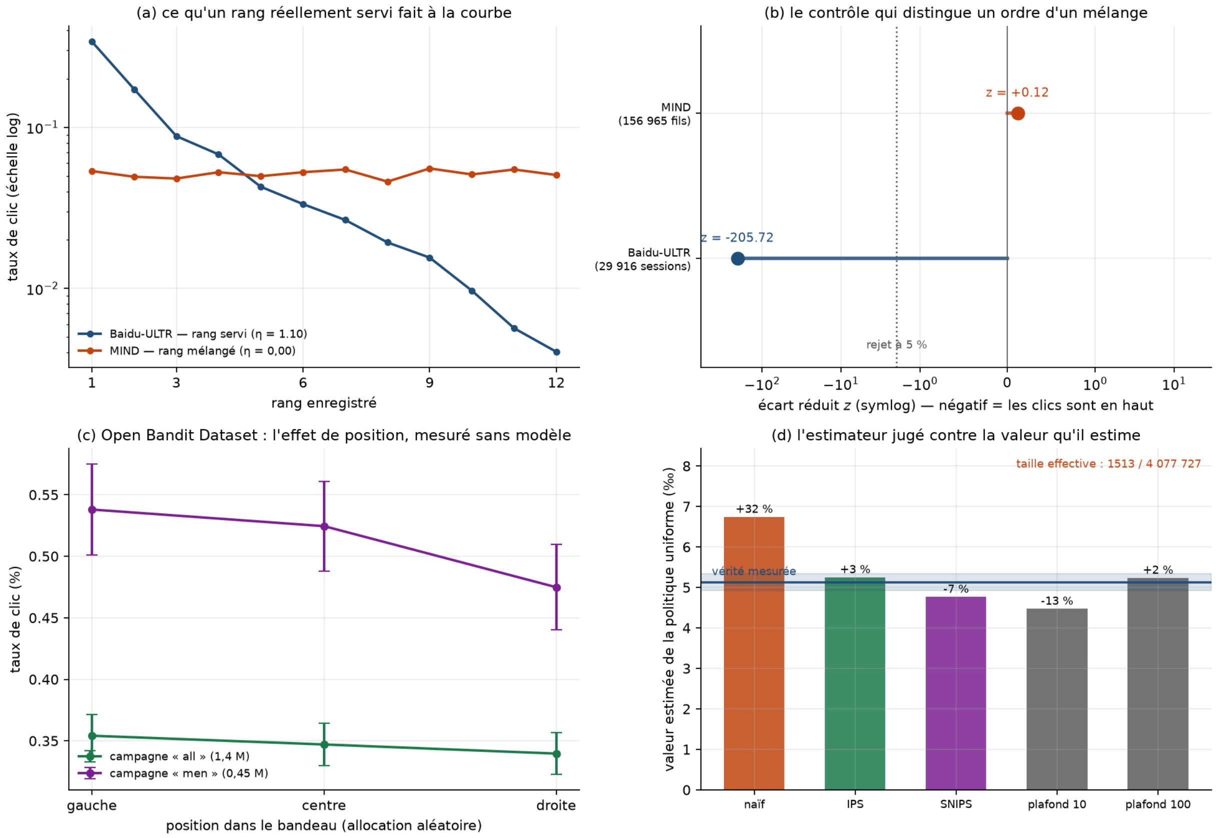

Figure 2: What a genuinely served rank does to the click-rate curve; the exchangeability test on MIND and on Baidu-ULTR; the position effect measured without a model on the Open Bandit Dataset's random bucket; the counterfactual estimator judged against the value it estimates.

at twenty, +0.255 at a hundred, each with a standard error below 0.007. **Five confident and incompatible figures from one dataset.**

Shuffling does not debias the clicks: they were produced under the real order and carry its bias in full. What the shuffle removes is the *rank*, the only regressor that would allow correction. On a simulated log of true severity 1.00, the estimate moves from 1.003 to -0.003 after shuffling, at identical clicks; evaluating a re-ranking becomes vacuous by construction.

6.2 Baidu-ULTR: the positive control

On a slice of 524,164 documents served across 64,200 search sessions [Zou et al., 2022], the same test rejects at $z = -205.7$ ($p < 10^{-12}$) and on the side theory prescribes. Severity is $\hat{\eta} = 1.10 \pm 0.09$ by document fixed effects, against 1.49 by aggregate fit — the gap being the quality confounder, the platform placing the best documents on top. Coverage remains thin: 55 documents carry the estimate, out of 444,709, because the same URL rarely reappears.

6.3 Open Bandit Dataset: position measured without a model

The random bucket allocates items to positions by chance: the click-rate difference between positions is a *causal* effect, with no examination model to posit. On a three-thumbnail horizontal banner it is $\hat{\eta} = 0.037$ ("all" campaign, 1.37 million impressions) to $\hat{\eta} = 0.105$ ("men" campaign).

η **is therefore a property of the surface, not a constant**: an order of magnitude separates a vertical results page from a three-thumbnail banner. Transporting one to the other is exactly the error whose cost §4 quantifies.

6.4 What these three logs do not permit

None carries both the *served rank* and an *interpretable viewpoint label*. MIND has editorial categories without the rank; Baidu-ULTR has the rank without labels; the Open Bandit Dataset has rank, propensity and three categorical attributes, but anonymised — and a diversity computed over hashes says nothing, the reference catalogue being a political declaration.

The index of §2 has therefore never been measured on a real feed. This is not a gap in method: it is a gap in data.

7 Limitations

These are stated here rather than left to reviewers.

7.1 On the index and the standard

The retained form has never been measured on a real feed. It is defined (2), its cost is quantified by exhaustive enumeration, and that is all. The gap is one of *data*, not of method (§6): it would take a log that close it.

The floor level cannot be deduced from the measurement. This work establishes the form of the standard and its price, not its value. Setting 0.60 rather than 0.40 is a political decision no computation presented here settles.

Discretisation into viewpoints is a political choice. Whoever defines the modalities defines the index. Measuring by divergence to a declared catalogue makes the choice explicit rather than burying it; it does not remove it.

The index remains gameable beyond what any automatic measure detects. The attacks of §3 are closed, but a platform can still serve formally divergent, substantively empty items — bin diversity without argument diversity. Any mandated metric becomes a target.

A floor is a constraint on what citizens see. Defensible, but an intervention in public debate, and presenting it as a neutral technical measure would be dishonest. Measuring individual feeds also raises a privacy question that this note does not answer, by aggregation.

7.2 On the exposure measurements

The form $e(R) = R^{-\eta}$ is posited. Only its severity is estimated. A cascade examination model, or one depending on the item, would produce a different rank dependence — which test (5) would detect just as well, since it only tests independence, but which the minimum detectable severity would no longer summarise.

Exhaustive enumeration bounds feed size. Eight positions over four viewpoints. Nothing guarantees that burial behaves the same on a fifty-item feed, and asserting it would require an approximate optimisation that would itself have to be shown not to invent the result.

The measured datasets are slices. A slice of Baidu-ULTR is not Baidu-ULTR; the Open Bandit Dataset measures a fashion recommendation on three horizontal thumbnails, and transporting its figures to a news feed is precisely the operation §6 warns against.

The ground-truth confrontation concerns a uniform target policy , the only one this dataset has a bucket for. Nothing guarantees importance weighting behaves as well for a target further from the logging policy — on the contrary, the effective sample size would be smaller still.

7.3 The two deepest assumptions, tested

Two assumptions run through this entire note without ever having been tested: that examination follows a power law, and that relevance is known. Neither is.

The exchangeability test survives a change of click model. Under a *cascade* model [Craswell et al., 2008] — the reader moves down the feed, stops on finding what they wanted, and otherwise continues with probability γ — test (5) rejects between $z = -208$ and $z = -241$ depending on γ, i.e. *more strongly* than on Baidu-ULTR. That is the result required: the test assumes nothing about the shape of attention, it tests only independence, and its negative verdict on MIND therefore holds under any model.

The power law does not survive. Under cascade, examination decays *geometrically* rather than polynomially. Simulation gives here the ground truth real data never provides — the positions actually examined — and the comparison is unequivocal: at rank twelve, the fitted $R^{-\hat{\eta}}$ overstates real exposure by a factor of 40 at $\gamma = 0.95$, 129 at $\gamma = 0.85$ and 4,110 at $\gamma = 0.60$. The fit nonetheless looks excellent, with a standard error of 0.04.

The consequence bears directly on §6: the severity $\hat{\eta} = 1.10 \pm 0.09$ measured on Baidu-ULTR means something *only if* examination there follows a power law — and Baidu-ULTR is a search results page, the cascade model's home ground. Severity must therefore be published **with its assumed shape**, and that shape remains to be tested. The exchangeability test says whether the order carries information; it does not say in what form.

And estimated relevance erases the tuned methods' advantage. The comparison in The re-ranker comparison, removed from this note, assumed relevance known. Having the methods rank on *perceived* relevance and judging them on the true one leaves their order unchanged — filter and MMR remain ahead — but their advantage over random selection falls from 16.4 points at zero noise to 5.8 points at a rank correlation of 0.50, and *inverts* below 0.36. The reason is mechanical: random selection is the only re-ranker that does not use relevance, hence the only one whose shortfall is invariant to noise.

At realistic relevance, the gap between re-rankers therefore matters far less than the quality of the underlying relevance engine — which reinforces, by a third route, the conclusion that this work's contribution is not its algorithm.

7.4 A third check: the shape of examination

The exchangeability test says whether a log's order carries information; it does not say in what form, and §7.3 showed the form decides everything. A third check is therefore required, and it rests on a *conditional independence* rather than on a shape: under a position model, examination of rank R depends only on R, so the click is independent of what happened above; under cascade, a click above suppresses examination below.

For each cell $s = (\text{item}, \text{rank})$, the n_s impressions split into n_{1s} preceded by a click and n_{0s} not, for k_s clicks in total. Under independence those clicks distribute as a draw without replacement:

$$\mathbb{E}[a_s] = \frac{k_s n_{1s}}{n_s}, \qquad \mathbb{V}[a_s] = \frac{k_s(n_s - k_s)\, n_{1s} n_{0s}}{n_s^2(n_s - 1)}. \tag{6}$$

This is the Mantel-Haenszel statistic, stratified by cell; conditioning removes item quality.

The check works. It never rejects under a position model ($|z| < 1$ for $\eta \in \{0, 0.5, 1, 2\}$) and rejects massively under cascade ($z = -97$ to -307).

But it does not settle what it was meant to settle. A reader who stops clicking once served — while still scanning the feed — produces the same signature: $z = -144$ under a *pure* position model with a one-click budget, against -179 under cascade. Nothing in the clicks distinguishes them, as expected: in both cases the log shows that after a click there is nothing. The difference is nonetheless decisive, since under a budget the item placed below a click is examined, and under cascade it is not.

On Baidu-ULTR, no cascade signature. The standardised deviation there is *positive* — $+5.6$ at the most permissive threshold, $+0.5$ to $+0.6$ at the strictest — whereas a cascade would make it negative. The sign is that of the heterogeneity confounder, which the test announced as its limit. Three readings remain open: examination there is close to a position model; a cascade exists but the confounder masks it; or coverage is too thin, 108 cells at the strictest threshold.

A protocol error, caught. Restricting the log to multi-click sessions — where a one-click budget is excluded by construction — looked like the obvious way to separate the two models, and gives $z = -8.2$ ($p = 2 \times 10^{-16}$) on Baidu-ULTR. Yet a feed's click count is a *collider* of its individual clicks: conditioning on it induces negative dependence between them, and therefore manufactures the sought signature. On a log simulated under a pure position model, the same restriction moves z from $+0.74$ to -45.7. What was measured was not a property of the log but the selection performed.

What would settle it. A measure of *examination* rather than of clicks — display time, scroll depth, items slipped off screen. Baidu-ULTR publishes those columns; this work has not read them.

7.5 Exposure measured rather than estimated

Baidu-ULTR publishes a `displayed_time` column — the display time of each document — which this work had not read. It measures directly what was *shown*, where everything above observed only what was *clicked*.

It lifts the ambiguity of §7.4. On clicks, cascade and budget are conflated; on examination, separation is total: $z = -158$ to -412 under cascade against $z = -1.03$ under budget, whatever the budget. Applied to Baidu-ULTR, the measure **refutes cascade**: examination below a click is there *more* frequent, not less — 0.221 against 0.136 at rank nine, $z = +8.4$ to $+10.9$ depending on threshold. This is the heterogeneity confounder, measured directly.

It revises the published figure. Exposure severity, measured by document fixed effects on display, is 0.882 ± 0.046 over **143** documents, against 1.085 ± 0.093 over 55 for the click-based estimate. The estimate **overstates decay by** 23%, and the gap is not imprecision but an identified confounder: a click is the product of examination and attractiveness, and attractiveness also declines with rank.

It calls into question the form used throughout. Fitted on the *measured* curve, the three candidate forms give $R^2 = 0.72$ for the power law — with 42% maximum error —, 0.96 for geometric decay and 0.99 for a *screen fold* model. The $R^{-\eta}$ law is the worst of the three, and it errs most where the standard is decided: at rank two it predicts 0.47 where measurement gives 0.88.

Reservation. A positive display time attests that the document was shown, not that it was looked at. The measure therefore overstates exposure, by an amount this work cannot quantify.

7.6 Format, return, and an explanation withdrawn

Two Baidu-ULTR columns remained unexploited: `slipoff_count_after_click`, which counts documents that slipped off screen after a click, and `media_type`, which distinguishes display formats.

Cascade refuted a second time. The first column is non-zero for 43.7% of clicked rows and 0.5% of the rest: after a click, the log explicitly records that documents scrolled past, hence that the reader came back. Under a strict cascade it would always be zero. The share moreover grows with the depth of the click — 0.37 at rank one, 0.59 at rank six. The refutation of §7.5, obtained by a statistical test, is thus confirmed by a recorded fact, which is an independent route.

Format alters exposure, and not through geometry. A rich format — answer box, image, table — is taller: 273 pixels median against 192. One would therefore expect a pushing-down effect. At equal rank *and* comparable cumulative height, a rich format above nonetheless removes 8.4 points of examination from what follows, median over 21 strata all of the same sign. The effect therefore does not run through space occupied, and the remaining explanation is *satisfaction*: an answer box answers the question, and the reader stops descending.

The consequence bears on any rank-only measure, including ours: exposure at rank R depends on the **format of what is above it**, and no law $e(R)$ can represent it since it does not take the page as an argument. The attention discount w_R of (1) is therefore not a property of the surface but of the **page served**: two feeds of identical composition, served with different formats, do not expose the same thing.

This conclusion is **withdrawn** by §7.7: the strata above hold rank and pixels fixed, but not the *item*. The paragraph is kept as published, its correction coming immediately after.

And one explanation in this note is withdrawn. §7.5 attributed the two-stage shape of the measured curve to a *screen fold*, invoking the `serp_height` column. That column allows the hypothesis to be tested, and it refutes it: rank explains examination better (McFadden R^2 0.082) than cumulative height above (0.057), and pixels add only 0.0007 once rank is known. The finding — the curve is not a power law — holds; the mechanical explanation falls.

7.7 The page effect, once composition is removed

The previous paragraph compared enriched pages to the others at equal rank and comparable height. It lacked one control: the **item**. An answer box does not appear at random — it answers a factual question, which does not invite the same reading as an exploratory search. The measured gap could therefore come from the format, or from what the format appears on.

A check that settles it. We simulate a log where the format has *no* effect on examination, but where enriched pages land on systematically less-consulted items. Stratifying on rank alone returns a risk ratio of 0.785 there — roughly the figure published above — when the truth is 1. The same protocol, matched, returns 1.005 (95% CI $[0.993, 1.016]$), and recovers 0.896 when an effect of 0.90 is simulated. Rank stratification therefore manufactures the result it was meant to measure; matching does not.

The measurement. On Baidu-ULTR, at **same document and same rank**, the risk ratio on examination is 0.940 (95% CI $[0.916, 0.964]$, 1,106 strata, 31,045 impressions). A second matching, by **query** and independent of the first, returns 0.942 (95% CI $[0.869, 1.022]$) on seven times fewer impressions. Rank by rank no trend emerges and every interval overlaps: the simplest surviving hypothesis is a constant factor.

What it costs in practice. A constant factor is a rescaling, to which a normalised index is insensitive; only rank 1, which never carries anything above it, keeps the effect from vanishing altogether. Over four thousand simulated feeds of nine ranks and four viewpoints, ignoring page composition moves the index by a median 0.0025, where the $1/R$ convention moves it by 0.0350 — **fourteen times more**. Same order on severity: $\eta = 0.876$ ignoring the page against 0.854 on the counterfactual bare page.

Consequence. The attention discount w_R of (1) becomes **portable** again, which was the real stake: were it page-dependent, no measurement could be written without describing every page served. The `format` column of for format remains useful — it allows this check to be reproduced elsewhere — but it no longer conditions the computation. What must be required of a log is first the displays column, without which exposure cannot be measured at all.

Reservations. The item-matched design retains only 1,106 strata out of 404,478: those are documents served at the same rank in pages of differing composition, and nothing guarantees they resemble the others. The query matching agrees but establishes nothing on its own, its interval containing 1. Finally, grouping `media_type` into "ordinary" versus "enriched" remains a choice of this work.

7.8 The index, measured on a real feed

Six sections of this note repeat the same sentence: *no public dataset carries both the served rank and an interpretable viewpoint label.* It is false, and its falsity is instructive.

A fourth log. EB-NeRD (*Ekstra Bladet*, RecSys Challenge 2024) gives for each impression the served list `article_ids_inview` and attaches every article to a declared section. Both columns are there. Yet the exchangeability test of §4 returns $z = +1.05$ ($p = 0.29$) over 232,887 feeds: the recorded order carries no position information. This is not a power failure — the severity this log would have detected is 0.0066, 134 times smaller than the one measured on Baidu-ULTR. `article_ids_inview` is a **served set**, not an ordered feed.

What becomes measurable. The rank-blind form, never computed until now outside a simulation. On the window §2 prescribes — the user-day — the index is 0.498 at the median, that is 5.07 sections equally served out of 26, with 13.7% of user-days below 0.40 and 50.7% below 0.50. This figure does not say whether that is little or much; it gives the order of magnitude missing from any discussion of a floor.

What remains of the exposed index. Composition being known and order not, the exposed index is not determined but **constrained**. Exact enumeration of every distinct ordering yields bounds of median width 0.104. Three verdicts become available, and their asymmetry is the result: at a floor of 0.40, 32.4% of feeds breach it *whatever* their ordering — an enforceable finding without the missing column — 7.9% are certainly compliant, and 59.7% remain undecidable. Without the rank one can convict; one almost never acquits.

Reservations. `category_str` is a *section*, not a viewpoint: the measurement bears on topical diversity exposed, the substitute already used on MIND. The severity $\eta = 0.88$ is transported from Baidu-ULTR, a search engine; §7.7 established that it transports from page to page, not from platform to platform, hence a sensitivity test that does not reverse the conclusion. The exact bounds cover only feeds of ten items or fewer, 64% of the log.

7.9 Whoever chooses the catalogue chooses the index

§2 has carried an unquantified caveat from the start: discretising into viewpoints is a political choice, and whoever defines the modalities defines the index. The user-days of §7.8 finally allow us to say *by how much.*

Three quantities, three behaviours. The **level** has no absolute meaning: the same corpus scores 0.498 over twenty-six sections and 0.924 over three. The **share below the floor** follows: at 0.40 it goes from 13.7% to 1.5% — the same threshold becomes decorative without a single impression changing. The **ordering** between readers does not move: rank concordance with the full catalogue is 1.000 at twelve modalities, 0.911 at six, then -0.09 at three. It does not degrade, it collapses.

Why. Rare sections serve almost nothing: the top eleven cover 99.8% of the corpus. Merging the other fifteen merges anything at all for only one user-day in a thousand, hence an unchanged ordering. Dropping to six forces a merge on 64.5% of user-days, to three on 94.8%.

Size does not specify the catalogue. Twenty random groupings into six classes return a median of 0.682 ± 0.093 and a concordance of 0.769 ± 0.098, against 0.860 and 0.911 for the frequency grouping. It is the *list* that must appear in a standard, not the number.

And exposed diversity does not exist in the singular. The corpus carries two labelling axes — thematic section and declared tone. Their concordance is $\rho = +0.155$: among the quarter most diverse in sections, 23.8% are also in the quarter most diverse in tone, where independence would give 25%. One must therefore say **which axis** is measured, and that choice weighs at least as much as the threshold.

Reservations. The coarse catalogues are obtained by grouping an existing catalogue, which is not a catalogue designed as such. The corpus is that of a single newspaper, and it is the shape of its distribution tail that explains the stability observed up to twelve modalities.

7.10 What the counter-expertise withdrew or restricted

The measuring instruments were confronted with the field's literature and with four counter-tests. Three results of this note come out modified.

A withdrawn conclusion. The comparison of the four diversity measures (§3) was carried out *at the same nominal floor*. Since the measures do not live on the same scale, that comparison is meaningless: at equal actually exposed diversity, the two retained measures cost the same, within 0.6% of the exact bound. The claim that proximity to a target "resists" burial better is a scale artefact. What makes the standard is the **level** required and **rank-awareness**, not the choice of measure.

A restricted conclusion. The figure "certified at 0.70, it exposes 0.36" assumes a $1/R$ attention discount, i.e. $\eta = 1$. Recomputed with the *measured* severity of a three-thumbnail banner ($\eta \approx 0.1$), the same feed exposes 0.747: burial disappears. At $\eta = 2$ it exposes only 0.157, and the loophole costs 2.7% instead of 20.8%. The attention discount must therefore be **measured on the surface**, and the stable price is that of the rank-aware standard — 20.8 to 24.1% depending on the surface.

A widened uncertainty. The estimator of §4 assumes a multiplicative click model. The literature documents an *affine* one, where readers click out of trust on irrelevant top-ranked items [Vardasbi et al., 2020], and proves that inverse propensity weighting cannot correct it. Under that model the estimated severity drifts by +12.8% without the standard error signalling it. Far less severe than positing η blindly, but the published interval is too narrow.

A confirmed recommendation. The doubly robust estimator does *worse* than plain weighting on the Open Bandit Dataset (−4.9% against +2.5%), and the effective sample size does not move: it depends only on the importance weights. No estimator replaces exploration.

And a caution from the literature. On Baidu-ULTR — the very dataset where this note measures $\hat{\eta} = 1.10$ — Hager et al. [2024] find that position-bias corrections improve click prediction *without* improving ranking quality as judged by expert annotators. The presence of the bias is confirmed by four concordant methods, which corroborates our figure; the next step does not follow mechanically.

7.11 On the model

Calibration barely begun. A single composite parameter is estimated J, T, γ and α taken separately have no estimation procedure. The most promising route would be inference of Kramers-Moyal coefficients on long opinion series, which would allow the drift to be *tested* rather than fitted.

The model's one proper prediction is not verified. The emotional-charge mechanism α predicts a difference between registers — amplification, persistence, switching rate. Four independent measurements, including a replication across 440 subjects and a blind double-coded annotation, find no trace of it. The apparent gap in the pilot corpus was a selection artefact.

An analogy is not an explanation. Nothing here demonstrates that opinions *obey* statistical mechanics; the work establishes that a formalism borrowed from physics *reproduces* some of its behaviours. On the founding analogy itself, the verdict is sharper still.

Individuals are not spins. Intentionality, irony and strategic anticonformism have no equivalent in a spin model; opinions are multidimensional [Deffuant et al., 2000]. Above all, the environment is not passive: an algorithm optimises a cost function, which makes it a strategic actor rather than a thermal bath. That is the analogy's weakest point — and it is also what makes a filter optimising the index conceivable: a cost function can be changed.

Scope of the simulations. The regimes explored were chosen for legibility. No systematic sensitivity study was carried out, system sizes are modest, and the conclusions of the removed model were qualitative: they concerned the existence of regimes and the direction of dependencies, never transferable numerical values.

7.12 On the annotation

The coders are not independent. Blind double coding of the corpus yields a Fleiss κ of 0.921, but the three coders are instances of the same language model: their agreement necessarily overstates what independent human judges would produce. It is the last methodological reservation that no computation lifts.

8 Reproducibility

The entire work is openly available and runs in containers, with 556 automated tests. The structural checks cover the entropy of a degenerate feed, severity estimated on simulated data of known severity, the exactness of the index bounds, and every figure published in this note, recomputed from the versioned digests.

The impression logs used are not redistributed — they weigh 135 MB to 2.2 GB and each carries its own licence — but the repository keeps verified digests of them, which reproduce every figure published here identically. Every figure is regenerated by a notebook.

`https://github.com/s-geffroy/Indice-Diversite-Exposee`
`https://s-geffroy.github.io/Indice-Diversite-Exposee/`

This work is open to critical review. The most useful feedback would concern the retained form of the index and the level of any floor, or the measurement instruments of §4 and §5, where a methodological error would be the most costly.

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